

Ibrahim Yacoub

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OBJECTIVE

Ambitious Eagle Scout and Mechanical Engineering undergraduate at Texas A&M University with experience using various CAD and CAM software through interning with a professional mechanical engineering team and leading a robotics mechanical design and construction department. Seeking an internship to apply my technical and soft skills in problem solving, mechanical design, and system integration while continuing to grow and learn from working in a professional environment.

EDUCATION

Texas A&M University
Mechanical Engineering

College Station, TX

EXPERIENCE

Si-Ware Systems

Mechanical Engineering Intern

Cairo, Egypt

September 2025 – August 2026

- Researched material properties, machining, and forming methods to utilize the knowledge in design decision making and constraints
- Researched and developed a training module covering Geometric Dimensioning and Tolerancing (GD&T)
- Designed optomechanical systems from client specifications
- Researched and developed a training module covering Ingress Protection (IP) rating standards and testing methodologies
- Developed CAD for an IP6X dust ingress compliance testing system to validate sealing
- Utilized CAD, CAM, and CNC machine to design and manufacture a mount for Specac ATR products compatible with company spectrometer

FTC Robotics

Build Lead

Austin, TX

August 2022 – May 2026

- Led robot design, assembly, and testing for the high school competition team
- Coordinated team members to strategize and execute game plans during competitions.
- Applied project management concepts to ensure efficient product completion within time constraints

HotHint

COO

Austin, TX

August 2024 – May 2025

- Led operational planning and management for a student-founded startup
- Managed product development while promoting team collaboration and productivity
- Utilized problem-solving skills to develop innovative solutions to real-world issues
- Presented business proposals and product concepts to professional audiences

PROJECTS

Structure Team – Marine Robotics

Galveston, TX

Unmanned Surface Vehicle

September 2026 – Present

- Designed a winch mechanism for an autonomous boat to test deep sea water
- Designed modular boat using T-slot beams to allow size and position adjustment

SKILLS

- Design, manufacturing, and construction of mechanical systems
- CAD: Onshape, Autodesk Inventor (Certified), Autodesk Fusion360, SolidWorks
- CAM: Autodesk Inventor and Autodesk Fusion360
- 3D printing and CNC milling
- GD&T Symbols and Sketching
- Java and Python
- Problem-solving and Leadership
- Arabic fluency